Problem Statement — Insurance Document Verification Portal

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1. Introduction

In modern insurance claims processing, document verification remains the single largest operational bottleneck. Annually, insurance providers process millions of customer-submitted documents—including hospital bills, death/disability certificates, motor surveyor inspection reports, police FIRs, and KYC identity proofs.

2. Problem Definition

There is currently no unified, lightweight, containerized verification platform available to insurance carriers that:

- Ingests multi-category claim documents and policy proofs in real time.
- Extracts and indexes key policy parameters (policy number, claimant identity, claimed amount).
- Automatically calculates heuristic authenticity scores and flags suspicious documents (risk score ≥ 70%).
- Provides a centralized, searchable registry with role-based workflow progression.
- Operates reliably within an automated CI/CD pipeline ensuring zero-downtime containerized updates.

3. Target Users

#	User Persona	Operational Role & Responsibilities
1	Policyholders / Claimants	Upload policy proofs and claim bills; track real-time verification status.
2	Underwriters / Verification Officers	Review queued documents, inspect metadata, update workflow status with notes.
3	Fraud Investigators (SIU)	Receive high-risk fraud alerts (risk score ≥ 70%) and quarantine fraudulent files.
4	DevOps Engineers	Automate CI/CD pipelines, container images, test suites, and target node provisioning.

4. Measurable Success Criteria

#	Criteria	Target Benchmark
1	Automated Intake & Triage Time	Under 2 seconds per document
2	Fraud Detection Accuracy	100% of documents with risk score ≥ 70% flagged
3	System Availability & Uptime	99.9% uptime during testing cycles
4	CI/CD Deployment Cycle	Under 3 minutes from commit to running container
5	Automated Test Coverage	100% pass rate across 6 critical user journeys